

SOHO Network

XYZ company is a fast-growing company in Eastern Australia with more than 2 million customers globally. The company deals with selling and buying of food items, which are basically operated from the headquarters. The company is intending to open a branch near the local village Bonalbo. Thus, the company requires young IT graduates to design the network for the branch. The network is intended to operate separately from the HQ network.

Being a small network, the company has the following requirements during implementation;

- a) One router and one switch to be used (all CISCO products).
- b) 3 departments (Admin/IT, Finance/HR and Customer service/Reception)
- c) Each department is required to be in different VLANS.
- d) Each department is required to have wireless network for the users.
- e) Host devices in the network are required to obtain IPv4 address automatically.
- f) Devices in all the departments are required to communicate with each other.

Assume the ISP gave out a base network of 192.168.1.0, you as the young network engineer who has been hired, design and implement a network considering the above requirements.

Base Network Address = 192.168.1.0

No. of required subnets = 3

Pattern is 2^n , $n=2$, $2^2=4$

N=2 (borrowed bits)

Class C => 255.255.255.0 => 11111111.11111111.11111111.00000000

After borrowing 2 bits,

11111111.11111111.11111111.11000000 => 255.255.255.192

Block size = 64 ($256\%4=64$)

First subnet,

Network = 192.168.1.0

Range = 192.168.1.1 - 192.168.1.62

Broadcast Address = 192.168.1.63

Subnet mask = 255.255.255.192

Second subnet,

Network = 192.168.1.64

Range = 192.168.1.65 - 192.168.1.126

Broadcast Address = 192.168.1.127

Subnet mask = 255.255.255.192

Third subnet,

Network = 192.168.1.128

Range = 192.168.1.129 - 192.168.1.190

Broadcast Address = 192.168.1.191

Subnet mask = 255.255.255.192

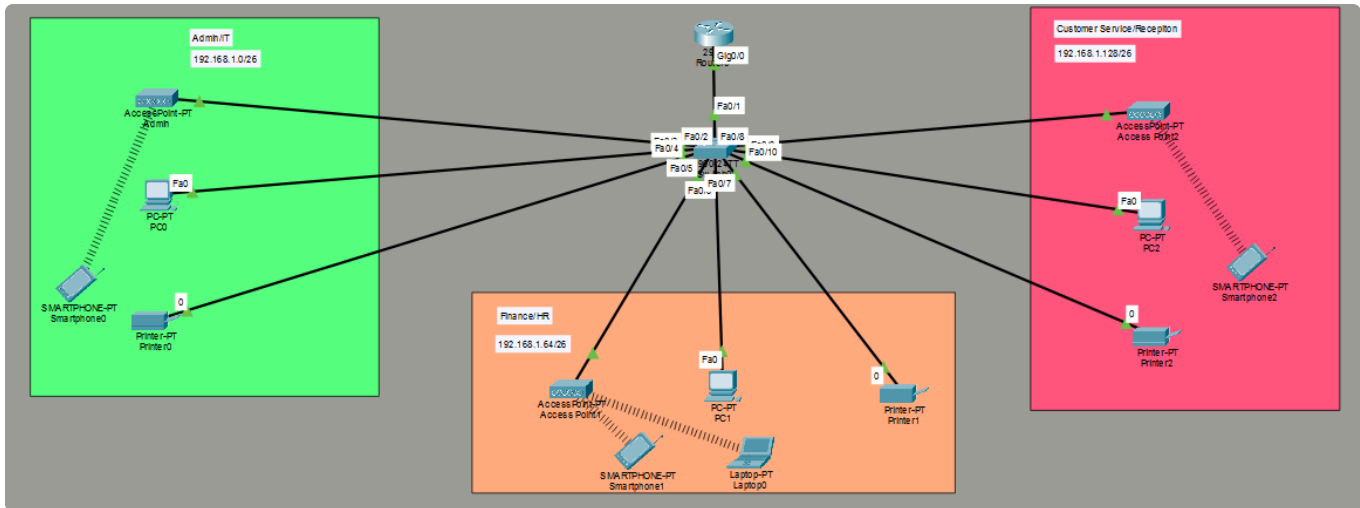
Fourth subnet,(we only need 3 subnets, so we don't use this)

Network = 192.168.1.192

Range = 192.168.1.193 - 192.168.1.254

Broadcast Address = 192.168.1.255

Subnet mask = 255.255.255.192



In switch

```
Switch(config)#int range fa0/2-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#do wr
```

```
Switch(config)#int range fa0/5-7
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#do wr
```

```
Switch(config)#int range fa0/8-10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#do wr
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#do wr
```

Configure access point

Goto port 1 =>rename access point name =>select authentication method=>set password

In Router

int gig0/0(interface that is connected to switch)

no shut

exit

```
Router(config)#int gig0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.1.1 255.255.255.192
Router(config-subif)#do wr
Router(config-subif)#exit
Router(config)#int gig0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.1.65 255.255.255.192
Router(config-subif)#do wr
Router(config-subif)#exit
Router(config)#int gig0/0.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.1.129 255.255.255.192
Router(config-subif)#do wr
```

```
Router(config)#service dhcp
Router(config)#ip dhcp pool Admin-pool
Router(dhcp-config)#network 192.168.1.0 255.255.255.192
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 192.168.1.1
Router(dhcp-config)#domain-name Admin.com
Router(dhcp-config)#exit
Router(config)#ip dhcp pool Finance-pool
Router(dhcp-config)#network 192.168.1.64 255.255.255.192
Router(dhcp-config)#default-router 192.168.1.65
Router(dhcp-config)#dns-server 192.168.1.65
Router(dhcp-config)#domain-name Finance.com
Router(dhcp-config)#exit
Router(config)#ip dhcp pool Customer-pool
Router(dhcp-config)#network 192.168.1.128 255.255.255.192
Router(dhcp-config)#default-router 192.168.1.129
Router(dhcp-config)#dns-server 192.168.1.129
Router(dhcp-config)#domain-name Customer.com
Router(dhcp-config)#exit
```

In smartphone

Goto wireless => fill SSID => select password method => fill password

For wireless connection we can use any wireless device such as smartphone, laptop, table (for laptop you need to add module like WPC300N for wireless connection)